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# EXPERIMENT NO - 1

## Title

**Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.**

## Experiment 1.1: Enjoying Sport

### Aim

To find the most specific hypothesis for whether a person enjoys sport based on weather conditions.

### Procedure

- Initialize the hypothesis with Ø for every attribute.
- Read each training example and separate its attributes from the target label.
- Ignore negative examples (target = No).
- For the first positive example, copy its attribute values into the hypothesis.
- For each later positive example, replace any differing attribute value with ?.
- Display the final hypothesis and interpret its constraints.

### Code

```
# FIND-S: Experiment 1: Enjoying Sport
training_data = [['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'], ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'], ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'], ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']]

num_attributes = len(training_data[0]) - 1
hypothesis = ["Ø"] * num_attributes

for example in training_data:
    attributes = example[:-1]
    target = example[-1]

    if target == "Yes":
        if hypothesis == ["Ø"] * num_attributes:
            hypothesis = attributes.copy()
        else:
            for i in range(num_attributes):
```

```
            if hypothesis[i] != attributes[i]:
                hypothesis[i] = "?"

print("Final hypothesis:", hypothesis)
```

## Input

Training data:

| Sky | Air Temp. | Humidity | Wind | Water | Forecast | Enjoy Sport? |
|---|---|---|---|---|---|---|
| Sunny | Warm | Normal | Strong | Warm | Same | Yes |
| Sunny | Warm | High | Strong | Warm | Same | Yes |
| Rainy | Cold | High | Strong | Warm | Change | No |
| Sunny | Warm | High | Strong | Cool | Change | Yes |

## Output

Final hypothesis: ['Sunny', 'Warm', '?', 'Strong', '?', '?']

## Result

The FIND-S algorithm was implemented for this dataset. The final hypothesis is ['Sunny', 'Warm', '?', 'Strong', '?', '?']. Sky must be Sunny, air temperature Warm, and wind Strong. Humidity, water temperature, and forecast can be any value. This result summarizes the positive training examples; it does not establish a causal rule or guarantee performance on unseen data.

## Experiment 1.2: Student Eligible for a Scholarship

### Aim

To generalize positive examples of students who qualify for a scholarship using academic and activity attributes.

### Procedure

- Initialize the hypothesis with Ø for every attribute.
- Read each training example and separate its attributes from the target label.
- Ignore negative examples (target = No).
- For the first positive example, copy its attribute values into the hypothesis.
- For each later positive example, replace any differing attribute value with ?.
- Display the final hypothesis and interpret its constraints.

### Code

```
# FIND-S: Experiment 2: Student Eligible for a Scholarship
training_data = [['High', 'Good', 'Completed', 'Yes', 'Low', 'Yes'], ['High', 'Good', 'Completed', 'No', 'Low', 'Yes'], ['Medium', 'Poor', 'Pending', 'No', 'High', 'No'], ['High', 'Good', 'Completed', 'Yes', 'Medium', 'Yes']]

num_attributes = len(training_data[0]) - 1
hypothesis = ["Ø"] * num_attributes

for example in training_data:
    attributes = example[:-1]
    target = example[-1]

    if target == "Yes":
        if hypothesis == ["Ø"] * num_attributes:
            hypothesis = attributes.copy()
        else:
            for i in range(num_attributes):
                if hypothesis[i] != attributes[i]:
                    hypothesis[i] = "?"

print("Final hypothesis:", hypothesis)
```

## Input

Training data:

| Marks | Attendance | Projects | Sports | Income | Eligible? |
|---|---|---|---|---|---|
| High | Good | Completed | Yes | Low | Yes |
| High | Good | Completed | No | Low | Yes |
| Medium | Poor | Pending | No | High | No |
| High | Good | Completed | Yes | Medium | Yes |

## Output

Final hypothesis: ['High', 'Good', 'Completed', '?', '?']

## Result

The FIND-S algorithm was implemented for this dataset. The final hypothesis is ['High', 'Good', 'Completed', '?', '?']. Marks must be High, attendance Good, and projects Completed. Sports participation and income may be any value among the observed examples. This result summarizes the positive training examples; it does not establish a causal rule or guarantee performance on unseen data.

## Experiment 1.3: Play Cricket

### Aim

To determine the most specific hypothesis for playing cricket based on outdoor conditions.

### Procedure

- Initialize the hypothesis with Ø for every attribute.
- Read each training example and separate its attributes from the target label.
- Ignore negative examples (target = No).
- For the first positive example, copy its attribute values into the hypothesis.
- For each later positive example, replace any differing attribute value with ?.
- Display the final hypothesis and interpret its constraints.

### Code

```
# FIND-S: Experiment 3: Play Cricket
training_data = [['Clear', 'Mild', 'Dry', 'Low', 'Good', 'Yes'], ['Clear', 'Warm', 'Dry', 'Low', 'Good',
'Yes'], ['Rainy', 'Cool', 'Wet', 'High', 'Poor', 'No'], ['Clear', 'Warm', 'Dry', 'High', 'Good', 'Yes']]

num_attributes = len(training_data[0]) - 1
hypothesis = ["Ø"] * num_attributes

for example in training_data:
    attributes = example[:-1]
    target = example[-1]

    if target == "Yes":
        if hypothesis == ["Ø"] * num_attributes:
            hypothesis = attributes.copy()
        else:
            for i in range(num_attributes):
                if hypothesis[i] != attributes[i]:
                    hypothesis[i] = "?"

print("Final hypothesis:", hypothesis)
```

## Input

Training data:

| Weather | Temperature | Ground | Wind | Visibility | Play Cricket? |
|---|---|---|---|---|---|
| Clear | Mild | Dry | Low | Good | Yes |
| Clear | Warm | Dry | Low | Good | Yes |
| Rainy | Cool | Wet | High | Poor | No |
| Clear | Warm | Dry | High | Good | Yes |

## Output

Final hypothesis: ['Clear', '?', 'Dry', '?', 'Good']

## Result

The FIND-S algorithm was implemented for this dataset. The final hypothesis is ['Clear', '?', 'Dry', '?', 'Good']. Weather must be Clear, the ground Dry, and visibility Good. Temperature and wind may be any value. This result summarizes the positive training examples; it does not establish a causal rule or guarantee performance on unseen data.

## Experiment 1.4: Approving a Loan

### Aim

To generalize positive examples of loan approval based on applicant attributes.

### Procedure

- Initialize the hypothesis with Ø for every attribute.
- Read each training example and separate its attributes from the target label.
- Ignore negative examples (target = No).
- For the first positive example, copy its attribute values into the hypothesis.
- For each later positive example, replace any differing attribute value with ?.
- Display the final hypothesis and interpret its constraints.

### Code

```
# FIND-S: Experiment 4: Approving a Loan
training_data = [['Good', 'Permanent', 'High', 'Low', 'Complete', 'Yes'], ['Good', 'Permanent', 'Medium', 'Low', 'Complete', 'Yes'], ['Poor', 'Temporary', 'Low', 'High', 'Incomplete', 'No'], ['Good', 'Permanent', 'High', 'Medium', 'Complete', 'Yes']]

num_attributes = len(training_data[0]) - 1
hypothesis = ["Ø"] * num_attributes

for example in training_data:
    attributes = example[:-1]
    target = example[-1]

    if target == "Yes":
        if hypothesis == ["Ø"] * num_attributes:
            hypothesis = attributes.copy()
        else:
            for i in range(num_attributes):
                if hypothesis[i] != attributes[i]:
                    hypothesis[i] = "?"

print("Final hypothesis:", hypothesis)
```

## Input

Training data:

| Credit Score | Employment | Income | Existing Debt | Documents | Loan Approved? |
|---|---|---|---|---|---|
| Good | Permanent | High | Low | Complete | Yes |
| Good | Permanent | Medium | Low | Complete | Yes |
| Poor | Temporary | Low | High | Incomplete | No |
| Good | Permanent | High | Medium | Complete | Yes |

## Output

Final hypothesis: ['Good', 'Permanent', '?', '?', 'Complete']

## Result

The FIND-S algorithm was implemented for this dataset. The final hypothesis is ['Good', 'Permanent', '?', '?', 'Complete']. Credit score must be Good, employment Permanent, and documents Complete. Income and existing debt generalize to any value in these examples. This result summarizes the positive training examples; it does not establish a causal rule or guarantee performance on unseen data.

## Experiment 1.5: Plant Growth Suitable

### Aim

To find a hypothesis describing conditions associated with suitable plant growth.

### Procedure

- Initialize the hypothesis with Ø for every attribute.
- Read each training example and separate its attributes from the target label.
- Ignore negative examples (target = No).
- For the first positive example, copy its attribute values into the hypothesis.
- For each later positive example, replace any differing attribute value with ?.
- Display the final hypothesis and interpret its constraints.

### Code

```
# FIND-S: Experiment 5: Plant Growth Suitable
training_data = [['Loamy', 'High', 'Moderate', 'Warm', 'Organic', 'Yes'], ['Loamy', 'High',
'Moderate', 'Mild', 'Organic', 'Yes'], ['Sandy', 'Low', 'Low', 'Cold', 'None', 'No'], ['Loamy', 'High',
'High', 'Warm', 'Organic', 'Yes']]

num_attributes = len(training_data[0]) - 1
hypothesis = ["Ø"] * num_attributes

for example in training_data:
    attributes = example[:-1]
    target = example[-1]

    if target == "Yes":
        if hypothesis == ["Ø"] * num_attributes:
            hypothesis = attributes.copy()
        else:
            for i in range(num_attributes):
                if hypothesis[i] != attributes[i]:
                    hypothesis[i] = "?"

print("Final hypothesis:", hypothesis)
```

## Input

Training data:

| Soil | Sunlight | Water | Temperature | Fertilizer | Good Growth? |
|---|---|---|---|---|---|
| Loamy | High | Moderate | Warm | Organic | Yes |
| Loamy | High | Moderate | Mild | Organic | Yes |
| Sandy | Low | Low | Cold | None | No |
| Loamy | High | High | Warm | Organic | Yes |

## Output

Final hypothesis: ['Loamy', 'High', '?', '?', 'Organic']

## Result

The FIND-S algorithm was implemented for this dataset. The final hypothesis is ['Loamy', 'High', '?', '?', 'Organic']. Soil must be Loamy, sunlight High, and fertilizer Organic. Water and temperature may be any value among the positive examples. This result summarizes the positive training examples; it does not establish a causal rule or guarantee performance on unseen data.